CS & DA

Probability and Statistics

Joint PMF & PDF

DPP- 01

Discussion Notes

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[NAT]

#Q. Consider two random variables X and Y with joint PMF given in Table.

X \ Y	Y =2	Y = 4	Y = 5	P(x)
X =1	1/12	1/24	1/24	= 4/24
X =2	1/6	1/12	1/8	= 9/24
X =3	1/4	1/8	1/12	= 11/24
P(Y)	6/12	6/24	6/24	1

X	1	2	3
P(X)	$\frac{1}{6}$	$\frac{3}{8}$	$\frac{11}{24}$

$\Sigma p_i = 1$

Y	2	4	5
P(Y)	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$

$\Sigma p_i = 1$

1) Find P (X ≤ 2, Y ≤ 4). $= P[X = 1 \text{ or } 2 \cap Y = 2 \text{ or } 4]$

$= P[(12) \text{ or } (14) \text{ or } (22) \text{ or } (24)]$

$= \frac{1}{12} + \frac{1}{24} + \frac{1}{6} + \frac{1}{12} = \frac{2+1+4+2}{24} = \frac{9}{24} = \frac{3}{8}$

[NAT]

2) Find the marginal PMFs of X and Y

$X =$

$P(X) =$

$Y:$

$P(Y):$

[NAT]

3) Find P (Y=2| X =1) $= \frac{P(Y=2 \cap X=1)}{P(X=1)} = \frac{P(1,2)}{P(X=1)} = \frac{1/12}{1/6} = \frac{1}{2}$

[NAT]

4) Are X and Y independent?

$$P(X=x_i \cap Y=y_j) = P(X=x_i)\cdot P(Y=y_j) \quad \forall\, i \,\&\, j$$

eg $P(X=2, Y=2) = P(2,2) = \frac{1}{6}$

$P(X=2)\cdot P(Y=2) = \frac{3}{8}\cdot\frac{1}{2} = \frac{3}{16} \neq \frac{1}{6}$ So Not Ind

[NAT]

#Q. For the adjoining bivariate probability distribution of X and Y , find:

X \ Y	Y =1	Y = 2	Y = 3	Y =4	Y =5	Y =6	P(X)
X =0	0	0	1/32	2/32	2/32	3/32	8/32
X =1	1/16	1/16	1/8	1/8	1/8	1/8	10/16
X =2	1/32	1/32	1/64	1/64	0	2/64	8/64
P(Y)	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{6}{32}$	$\frac{16}{64}$	

1) $P(X \leq 1, Y=2) = P(X = 0 \text{ or } 1 \cap Y = 2)$

$= P[(0,2) \text{ or } (1,2)]$

$= 0 + \frac{1}{16} = \frac{1}{16}$

2) $P(X \leq 1)$. $= P(x=0 \text{ or } 1) = \frac{8}{32} + \frac{10}{16} = \frac{8}{32} + \frac{20}{32} = \frac{28}{32}$

3) $P(Y \leq 3)$. $= P(Y = 1 \text{ or } 2 \text{ or } 3) = \frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{6+6+11}{64} = \frac{23}{64}$

y:	1	2	3	4	5	6
$p(y)$:	$\frac{3}{32}$	$\frac{3}{32}$	$\frac{11}{64}$	$\frac{13}{64}$	$\frac{6}{32}$	$\frac{16}{64}$

4) $P(X<3, Y \leq 4)$. $= P(X = 0 \text{ or } 1 \text{ or } 2 \cap Y = 1 \text{ or } 2 \text{ or } 3 \text{ or } 4)$

$$= P(X=0, Y \leq 4) + P(X=1, Y \leq 4) + P(X=2, Y \leq 4)$$

$$= \left(0 + 0 + \frac{1}{32} + \frac{2}{32}\right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{8} + \frac{1}{8}\right) + \left(\frac{1}{32} + \frac{1}{32} + \frac{1}{64} + \frac{1}{64}\right) = \frac{9}{16}$$

[NAT]

#Q. For the joint probability distribution of two random variables X and Y, find:

X \ Y	Y =1	Y = 2	Y = 3	Y =4	Total
X =1	4/36	3/36	2/36	1/36	10/36
X =2	1/36	3/36	3/36	2/36	9/36
X =3	5/36	1/36	1/36	1/36	8/36
X =4	1/36	2/36	1/36	5/36	9/36
Total	11/36	9/36	7/36	9/36	1

X:	1	2	3	4
P(X):	$\frac{10}{36}$	$\frac{9}{36}$	$\frac{8}{36}$	$\frac{9}{36}$

Y:	1	2	3	4
P(Y):	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{9}{36}$

1)Find the marginal distributions of X amd Y

2) Conditional distribution of X given the value of Y =1 and that of Y given the value of X =2

X	1	2	3	4
$P(X=x/Y=1)$	$\frac{4}{11}$	$\frac{1}{11}$	$\frac{5}{11}$	$\frac{1}{11}$

$$P(X=1/Y=1)=\frac{P(1,1)}{P(Y=1)}=\frac{4/36}{11/36}=\frac{4}{11}$$

$$P(X=2/Y=1)=\frac{P(2,1)}{11/36}=\frac{1/36}{11/36}=\frac{1}{11}$$

$$P(X=3/Y=1)=\frac{P(3,1)}{11/36}=\frac{5/36}{11/36}=\frac{5}{11}$$

$$P(X=4/Y=1)=\frac{1/36}{11/36}$$

Y	1	2	3	4
$P(Y=y/X=2)$	$\frac{1}{9}$	$\frac{3}{9}$	$\frac{3}{9}$	$\frac{2}{9}$

$$P(Y=1/X=2)=\frac{P(2,1)}{P(X=2)}=\frac{1/36}{9/36}=\frac{1}{9}$$

$$P(Y=2/X=2)=\frac{P(2,2)}{P(X=2)}=\frac{3/36}{9/36}=3/9$$

$$P(Y=3/X=2)=\frac{P(2,3)}{P(X=2)}=\frac{3/36}{9/36}=\frac{3}{9}$$

$$P(Y=4/X=2)=\frac{P(2,4)}{P(X=2)}=\frac{2/36}{9/36}=\frac{2}{9}$$

[NAT]

#Q. I have a bag containing 40 blue marbles and 60 red marbles. I choose 10 marbles (without replacement) at random. Let X be the number of blue marbles and y be the number of red marbles. Find the joint PMF of X and Y.

eg 3 Blue & 4 Red

& we want to take 3 Marbles w/o Replacement, then Prob of drawing 1B & 2R are ?

M-I (BRR) or (RBR) or (RRB)

$$\left(\frac{3}{7}\times\frac{4}{6}\times\frac{3}{5}\right)+\left(\frac{4}{7}\times\frac{3}{6}\times\frac{3}{5}\right)+\left(\frac{4}{7}\times\frac{3}{6}\times\frac{3}{5}\right)=\left(\frac{4\times3\times3}{7\times6\times5}\right)3$$

M-II

$$\text{Req Prob}=\frac{{}^3C_1\times{}^4C_2}{{}^7C_3}=\frac{3\times\left(\frac{4\times3}{2}\right)}{\frac{7\times6\times5}{3\times2}}=\left(\frac{3\times4\times3}{7\times6\times5}\right)\times3$$

Fav Cases = {(0,10), (1,9), (2,8) - - (10,0)}

$x+y=10$.

$$P(x=x_i, y=y_j)=\begin{cases}\dfrac{{}^{40}C_{x_i}\ {}^{60}C_{y_j}}{{}^{100}C_{10}}, & x_i+y_j=10\\ 0, & \text{otherwise}\end{cases}$$

eg

$$P(x=1, y=2)=\frac{{}^3C_1\times{}^4C_2}{{}^7C_3}$$

[NAT]

#Q. The joint p.d.f of two random variables X and Y is given by:

$$f(x,y) = 9(1+x+y)/2((1+x)^4(1+y)^4) \quad 0 \le x < \infty, 0 < y < \infty$$

$$f(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_0^{\infty} \frac{9(1+x+y)}{2(1+x)^4(1+y)^4}\,dy = \frac{9}{2(1+x)^4}\int_0^{\infty}\frac{(1+y+x)}{(1+y)^4}\,dy = \frac{9}{2(1+x)^4}\int_0^{\infty}\left[(1+y)^{-3}+x(1+y)^{-4}\right]dy$$

Find the marginal distributions of X and Y , and the conditional distribution of Y for X=x.

$$= \frac{9}{2(1+x)^4}\left[\frac{(1+y)^{-2}}{-2} + x\cdot\frac{(1+y)^{-5}}{-5}\right]_0^{\infty} = \frac{9}{2(1+x)^4}\left[\frac{-1}{2(1+y)^2} - \frac{x}{5(1+y)^5}\right]_0^{\infty} = \frac{-9}{2(1+x)^4}\left[0-\left\{\frac{1}{2}+\frac{x}{5}\right\}\right]$$

$$= \frac{3}{4}\cdot\frac{(3+2x)}{(1+x)^4}, \quad 0 \le x < \infty$$

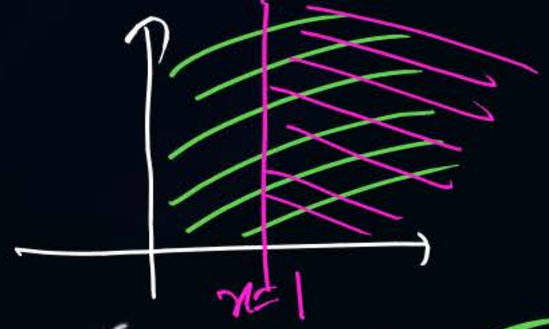
Similarly $f(y) = \int_0^{\infty} f(x,y)\,dx = \dots = \frac{3}{4}\cdot\frac{(3+2y)}{(1+y)^4}, \quad 0 \le y < \infty$

$$\boxed{P\left[Y=y \,/\, X=x\right] = \frac{f(x,y)}{f(x)}} = \frac{\dfrac{9(1+x+y)}{2(1+x)^4(1+y)^4}}{\dfrac{3}{4}\,\dfrac{(3+2x)}{(1+x)^4}} = \frac{6(1+x+y)}{(1+y)^4(3+2x)}$$

$$0 \le y < \infty$$

[NAT]

#Q. Given-



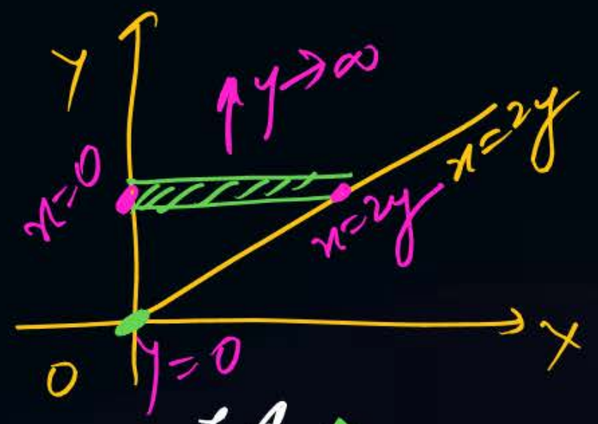
$$f(x,y) = \left\{ (e^{-(x+y)}), 0 \le x < \infty, 0 \le y < \infty \right.$$

Calculate the value of 1) P(X>1) $= \int_{y=0}^{\infty}\int_{1}^{\infty} f(x,y)\,dx\,dy = \int_{y=0}^{\infty}\int_{x=1}^{\infty} e^{-x}\cdot e^{-y}\,dx\,dy$

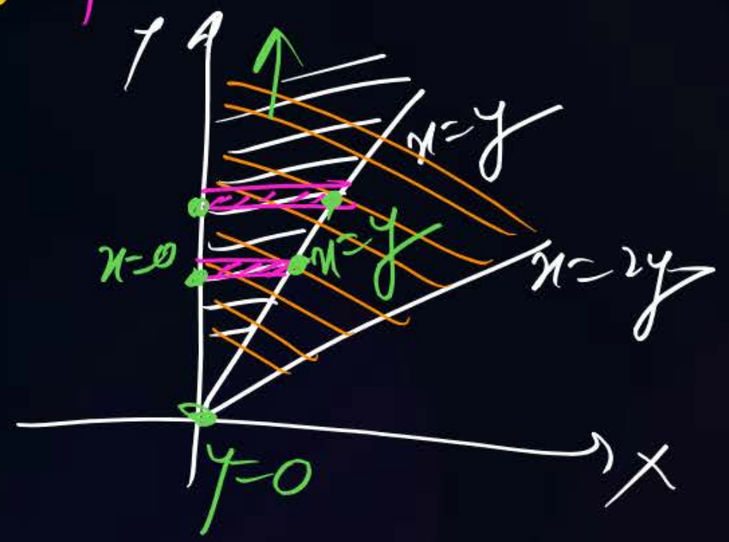
$$= \int_{x=1}^{\infty} e^{-x}\,dx \int_{y=0}^{\infty} e^{-y}\,dy = \frac{1}{e}\times 1 = e^{-1}$$

[NAT]

2) $P(X<Y|X<2Y) = \dfrac{P[x<y \cap x<2y]}{P(x<2y)} = \dfrac{P(x<y)}{\int_{x=0}^{\infty}\int_{y=0}^{2y} f(x,y)\,dx\,dy} = \dfrac{\int_{y=0}^{\infty}\int_{x=0}^{y} e^{-(x+y)}\,dx\,dy}{\int_{y=0}^{\infty}\int_{x=0}^{2y} e^{-(x+y)}\,dx\,dy}$



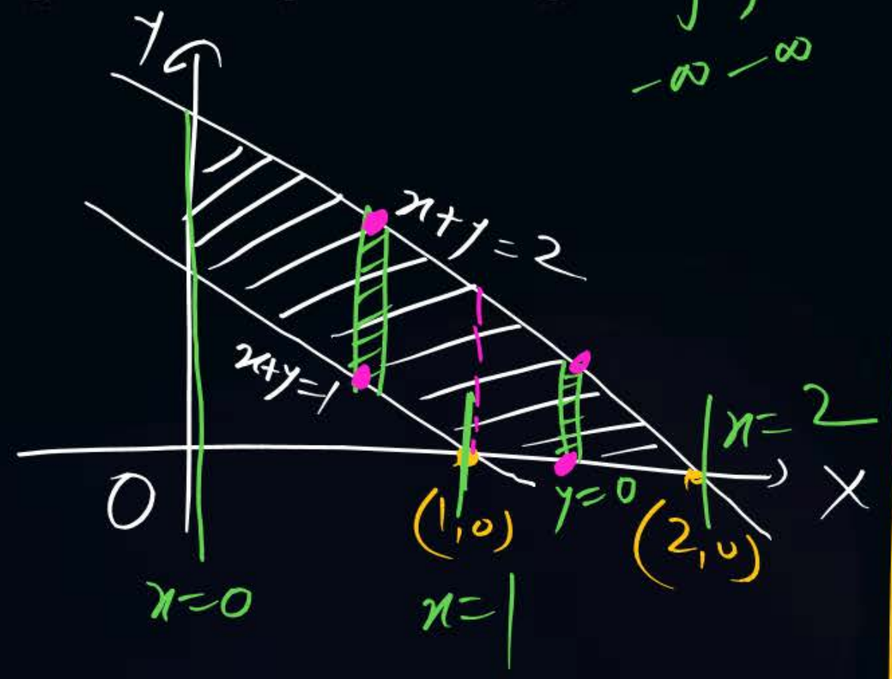
$$= \frac{\int_{y=0}^{\infty}\left(\frac{e^{-x}}{-1}\right)_{x=0}^{y}\left(e^{-y}dy\right)}{\int_{y=0}^{\infty}\left(\frac{e^{-x}}{-1}\right)_{x=0}^{2y}\left(e^{-y}dy\right)} = \frac{\int_0^{\infty}\left(1-e^{-y}\right)e^{-y}\,dy}{\int_0^{\infty}\left(1-e^{-2y}\right)e^{-y}\,dy} = \frac{\int_0^{\infty}\left(e^{-y}-e^{-2y}\right)dy}{\int_0^{\infty}\left(e^{-y}-e^{-3y}\right)dy}$$



$$= \frac{1/2}{2/3} = \frac{3}{4} \text{ Ans}$$

[NAT]

3) $P(1<X+Y<2) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = \int_{x=0}^{1}\int_{y=1-x}^{2-x} e^{-(x+y)}\,dy\,dx + \int_{x=1}^{2}\int_{y=0}^{2-x} e^{-(x+y)}\,dy\,dx$



$$= \int_{x=0}^{1}\left(\frac{e^{-y}}{-1}\right)_{y=1-x}^{2-x}\left(e^{-x}dx\right) + \int_{x=1}^{2}\left(\frac{e^{-y}}{-1}\right)_{0}^{2-x}\left(e^{-x}dx\right)$$

$$= \int_{0}^{1}\left(e^{x-1}-e^{x-2}\right)e^{-x}dx + \int_{x=1}^{2}\left(1-e^{x-2}\right)e^{-x}dx$$

$$= \int_{0}^{1}\left(e^{-1}-e^{-2}\right)dx + \int_{1}^{2}\left(e^{-x}-e^{-2}\right)dx$$

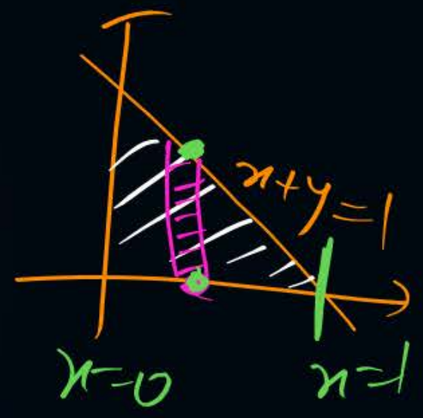
$$= \frac{2}{e}-\frac{3}{e^2}$$

[NAT]

#Q. Let X and Y be jointly continuous random variables with joint PDF

$$f_{X,Y}(x,y)=\begin{cases} cx+1 & x,y\geq 0, x+y<1 \\ 0 & otherwise \end{cases}$$

1) Find the constant c.



w.k. that $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$

$$\int_{x=0}^{1}\int_{y=0}^{y=1-x} (cx+1)\,dy\,dx = 1$$

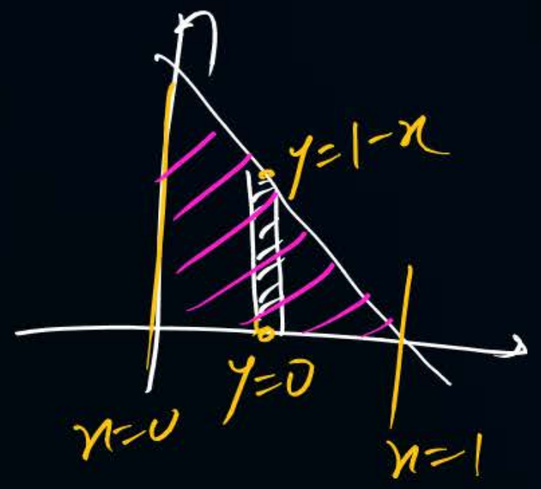
$$\int_{x=0}^{1} (1-x)(cx+1)\,dx = 1$$

$$c=3$$

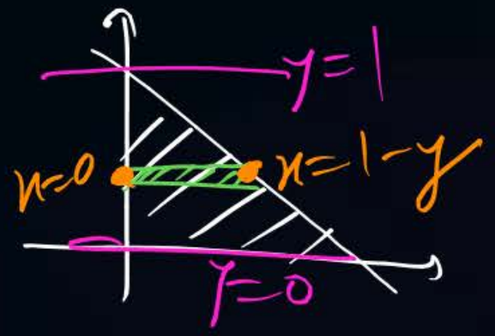
[NAT]

2) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.

$$f(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_{y=0}^{1-x} (3x+1)\,dy = \boxed{(3x+1)(1-x)} \quad 0 \le x \le 1$$



(ii)



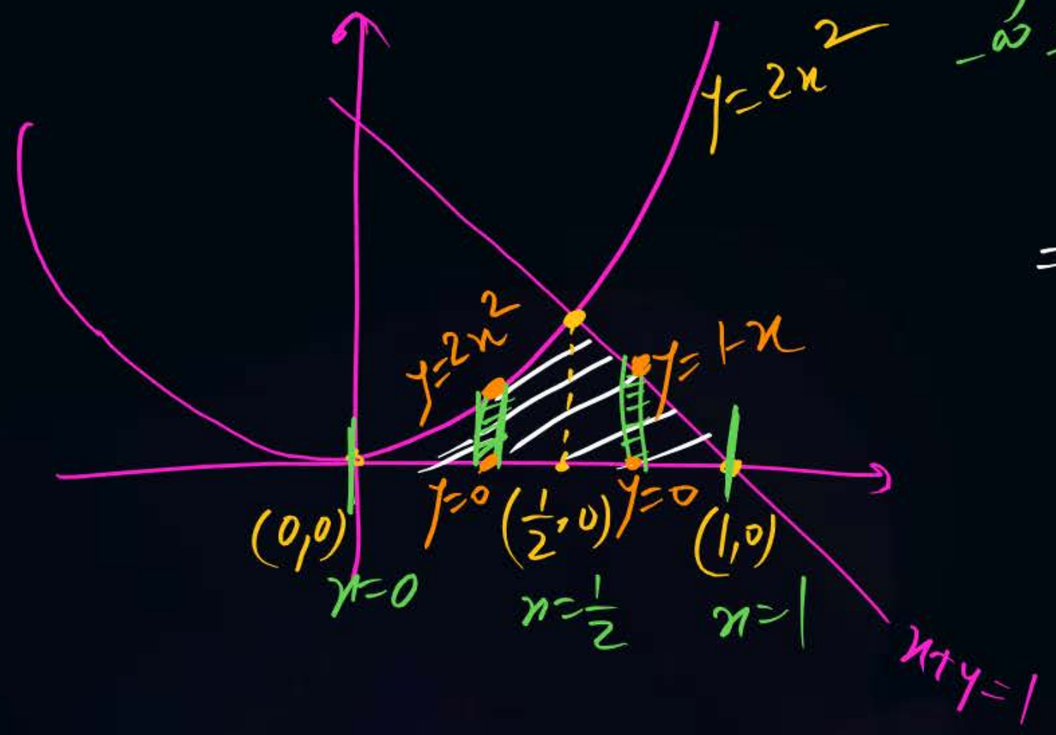
$$f(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_{x=0}^{1-y} (3x+1)\,dx = \left[3\left(\frac{x^2}{2}\right) + x\right]_0^{1-y} = \frac{3}{2}(1-y)^2 + (1-y)$$

$$= \boxed{\frac{1}{2}(1-y)(5-3y)}$$

$$0 \le y < \infty$$

[NAT]

3) Find $P(Y < 2X^2)$.



$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(3x+1)\,dx\,dy = \int_{x=0}^{\frac{1}{2}}\int_{y=0}^{2x}(3x+1)\,dy\,dx + \int_{x=\frac{1}{2}}^{1}\int_{y=0}^{1-x}(3x+1)\,dy\,dx$$

$$= \int_{x=0}^{\frac{1}{2}}(3x+1)(2x)\,dx + \int_{x=\frac{1}{2}}^{1}(3x+1)(1-x)\,dx$$

$$= 53/96$$

[NAT]

#Q. Let X and Y be jointly continuous random variables with joint PDF

$$f_{X,Y}(x,y)=\begin{cases}6e^{-(2x+3y)} & x,y\geq 0\\ 0 & otherwise\end{cases}$$

$f(x,y)=6e^{-2x}\cdot e^{-3y}$

$=(2e^{-2x})\cdot(3e^{-3y})$

$=f(x)\,f(y)$

1) Are X and Y independent? YES

ie these are Ind

E-Dist: $f(t)=\begin{cases}\mu e^{-\mu t}, & t\geq 0\\ 0, & \text{otw}\end{cases}$

$\Rightarrow \mu=3$ & $E(t)=\frac{1}{\mu}=\frac{1}{3}$

Cross check: $f(x)=\int_{-\infty}^{\infty}f(x,y)\,dy=\int_{0}^{\infty}6e^{-2x}\cdot e^{-3y}\,dy$

$=2e^{-2x}$ //

& $\boxed{f(y)=3e^{-3y}}$, $y\geq 0$

[NAT]

2) Find E [Y | [X >2].] $= E(Y) = \frac{1}{3}$

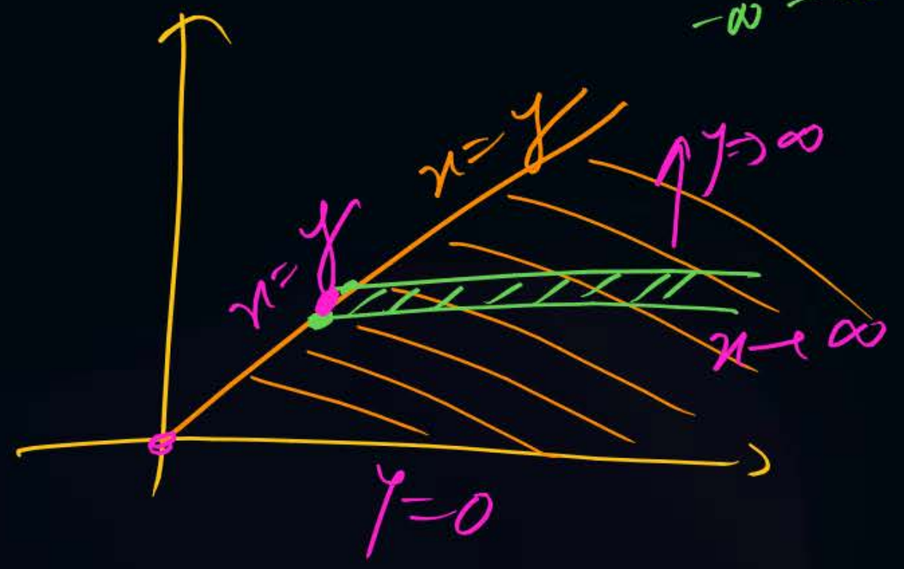
w. k that In Case of Independency, Condition has No significance.

Similarly $E(X) = \frac{1}{2}$

[NAT]

3) Find P (X >Y).

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = \int_{y=0}^{\infty}\int_{x=y}^{\infty}\left(6e^{-2x}\cdot e^{-3y}\right)dx\,dy$$



$$= \int_{y=0}^{\infty}\left(\frac{e^{-2x}}{-2}\right)_{y}^{\infty}\left(6e^{-3y}\,dy\right)$$

$$= \int_{0}^{\infty}\frac{e^{-2y}}{2}\cdot 6e^{-3y}\,dy = 3\int_{0}^{\infty}e^{-5y}\,dy = \frac{3}{5}$$

[NAT]

#Q. Let X be a continuous random radiable with PDF

$$f_X(x) = \begin{cases} 2x & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$$

$$f(y/x) = \begin{cases} \frac{1}{x-(-x)}, & -x \le y \le x \\ 0, & otw \end{cases}$$

Given X = x, the random variable Y is uniformly distributed on [−x, x].

1) Find the joint PDF F_{XY} (x,y)

ie $f(x,y) = \begin{cases} 1, & 0 \le x \le 1 \,\&\, -x \le y \le x \\ 0, & otw \end{cases}$

w.k.that, $\boxed{P(A \cap B) = P(A/B) \cdot P(B)}$

$\Rightarrow$ $f(x,y) = f(y/x) \cdot f(x)$

$= \frac{1}{2x} \cdot 2x = 1$

$$\boxed{f(x,y) = \begin{cases} 1, & |y| \le x \le 1 \\ 0, & otw \end{cases}}$$

[NAT]

2) Find $f_Y(y)$. $= \int_{-\infty}^{\infty} f(x,y)\,dx = \int_{x=|y|}^{1} 1\,dx = 1-|y|$

So $f(y) = \begin{cases} 1-|y|, & -1 \le y \le 1 \\ 0, & \text{otw} \end{cases}$

[NAT]

#Q. A bank operates a drive-up and a walk-up window. Here
X= the proportion of time the drive-up facility is in use
Y= the proportion of time the walk-up window in use
Say the manager has given the joint pdf based on his experience:

$$f(x,y) = \begin{cases} 6/5(x+y^2) & 0 \le x \le 1, \; 0 \le y \le \frac{1}{4} \\ 0 & otherwise \end{cases}$$

The probability that neither facility is busy more than one-quarter of the time is?

$$P\left(0<x<\frac{1}{4}, 0<y<\frac{1}{4}\right) = \int_0^{1/4}\int_0^{1/4} f(x,y)\,dx\,dy = \int_{y=0}^{1/4}\int_{x=0}^{1/4} \frac{6}{5}(x+y^2)\,dx\,dy$$

$$= \frac{6}{5}\int_{y=0}^{1/4}\left(\frac{x^2}{2}+xy^2\right)_{x=0}^{1/4} dy = \frac{6}{5}\int_{y=0}^{1/4}\left(\frac{1}{32}+\frac{y^2}{4}\right)dy = \frac{7}{640}$$

THANK - YOU